

Hybrid Model Predictive and Nonlinear Feedback Control for Chaos Suppression and MPPT Enhancement in DFIG-Based Wind Energy Systems

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Abstract—Doubly fed induction generator (DFIG)-based wind turbines are strongly nonlinear electromechanical systems that may exhibit irregular, and even chaotic, oscillations under abnormal operating conditions. This paper revisits the third-order rotor-current/speed model of the DFIG with a physically consistent parameter set, derives an equivalent dimensionless form, and shows by Lyapunov exponents and bifurcation analysis that chaotic motion appears only for strongly detuned mechanical damping and abnormal rotor excitation. A hybrid controller is then proposed in which nonlinear feedback linearization cancels the coupled dynamics and a constrained, receding-horizon model predictive controller (MPC) with integral action handles actuator limits; the MPC terminal weight and local gain are obtained from a linear matrix inequality (LMI) that guarantees nominal closed-loop stability. The controller is evaluated on chaos suppression and on maximum power point tracking (MPPT) of a 2-MW-class DFIG under turbulent wind and $\pm 20\%$ parameter mismatch, and compared with a feedback-linearization-only law and a PI law using the same gains. It suppresses chaos without steady-state error in all mismatch cases, reduces the recovery integral absolute error by 19–47% with respect to PI, and tracks the MPPT power reference with a root-mean-square error below 0.9% of rated power, five to eight times lower than with PI.

Index Terms—Doubly fed induction generator (DFIG), wind energy, chaotic dynamics, maximum power point tracking (MPPT), model predictive control (MPC), feedback linearization, linear matrix inequality (LMI).

I. INTRODUCTION

Rising electricity demand and the depletion of fossil resources call for sustainable energy sources such as wind [1]–[3]. Wind power has grown rapidly [4]: a record 117 GW was installed in 2024, bringing the cumulative capacity to about 1136 GW [5], driven by supportive policies and declining costs [6].

The doubly fed induction generator (DFIG) dominates large onshore wind farms because its partial-scale back-to-back converter, rated at about 25–30% of the generator power, allows decoupled control of active and reactive power over a wide speed range [7]–[11]. Its dynamic modelling, vector control and fault ride-through have been studied extensively [12]–[16], and grid codes impose strict requirements on its dynamic behaviour [17]–[19].

The electromechanical coupling between the rotor currents, the stator flux and the shaft speed makes the DFIG a strongly nonlinear system. Oscillatory transients [20] and, under particular parameter ranges or working conditions, chaotic behaviour have been reported [21], [22]; chaotic oscillations have also been analysed in other wind-turbine subsystems [23]. Chaos in electrical machines is usually analysed on dimensionless models [24], [25] and suppressed by nonlinear or adaptive feedback [24], [26], [27] or by predictive control [25], [28]–[31]; robust sliding-mode and fault-tolerant controllers have also been proposed for wind turbines [32], [33], and chaos synchronization has applications such as secure communications [34]. For MPPT, combining feedback linearization with MPC has proven effective [35].

A critical point, often overlooked, is whether the parameters used to exhibit chaos are physically admissible. In this paper: (i) the DFIG model of [36] is written with a physically consistent parameter set ($L_s, L_r > L_m$, $0 < \sigma < 1$) and all constants of the state equations are given explicitly; (ii) a dimensionless form is derived, which shows that chaos requires strongly abnormal damping and rotor excitation, and the chaotic regime is characterized by Lyapunov exponents and a bifurcation diagram; (iii) a hybrid controller combining feedback linearization with a constrained MPC is designed, with an LMI-based terminal cost that guarantees nominal stability; (iv) quantitative comparisons with two baselines under parameter mismatch are reported for chaos suppression and for MPPT tracking.

II. WIND TURBINE MODEL

The aerodynamic power captured by the rotor is [32], [33], [37]

$$P_a = \frac{1}{2} \rho \pi R^2 C_p(\lambda) v^3, \quad \lambda = \frac{R \Omega_t}{v}, \quad (1)$$

where ρ is the air density, R the blade radius, v the wind speed, Ω_t the turbine speed, and C_p the power coefficient, maximal ($C_{p,\max}$) at $\lambda = \lambda_{opt}$. We use $C_p(\lambda) = 0.5176(116/\lambda_i - 5)e^{-21/\lambda_i} + 0.0068\lambda$ with $1/\lambda_i = 1/\lambda - 0.035$ (zero pitch). Through a gearbox of ratio G , the generator mechanical speed is $\Omega_m = G\Omega_t$ and the electrical rotor speed is $\omega_r = n_p\Omega_m$,

n_p being the number of pole pairs. With the motor sign convention, the shaft dynamics referred to the generator side are

$$J \frac{d\Omega_m}{dt} = T_e - T_L - D \Omega_m, \quad T_L = -\frac{P_a}{\Omega_m}, \quad (2)$$

where J is the total equivalent inertia, D the viscous damping, and T_e the electromagnetic torque ($T_e < 0$ in generating mode). In the synchronous dq frame, $T_e = \frac{3}{2} n_p (\psi_{ds} i_{qs} - \psi_{qs} i_{ds})$.

III. DFIG MODEL AND CHAOTIC DYNAMICS

A. Physically Consistent State Model

The stator flux is aligned with the q axis ($\psi_{qs} = \psi_s$, $\psi_{ds} = 0$) and, neglecting the stator resistance in the flux relation, $u_{ds} = -V_s$, $u_{qs} = 0$, $\psi_s = V_s/\omega_s$, where V_s is the stator-voltage amplitude and ω_s the grid angular frequency. Then $i_{ds} = -(L_m/L_s) i_{dr}$ and $T_e = \frac{3}{2} n_p (L_m/L_s) \psi_s i_{dr}$. With $x = [x_1, x_2, x_3]^T = [i_{dr}, i_{qr}, \omega_r]^T$, the DFIG dynamics [36] read

$$\begin{cases} \dot{x}_1 = a_1 x_1 + (\omega_s - x_3) x_2 + a_2 x_3 + c_1 + u_1, \\ \dot{x}_2 = -(\omega_s - x_3) x_1 + a_1 x_2 + c_2 + u_2, \\ \dot{x}_3 = a_6 x_1 - a_7 x_3 - a_8 + u_3, \end{cases} \quad (3)$$

with $\sigma = 1 - L_m^2/(L_s L_r)$ and

$$\begin{aligned} a_1 &= -\frac{R_s L_m^2}{\sigma L_s^2 L_r} - \frac{R_r}{\sigma L_r}, & a_2 &= -\frac{L_m \psi_s}{\sigma L_r L_s}, \\ a_3 &= -\frac{L_m}{\sigma L_s L_r}, & a_4 &= \frac{1}{\sigma L_r}, & a_5 &= \frac{R_s L_m \psi_s}{\sigma L_s^2 L_r}, \\ a_6 &= \frac{3 n_p^2 L_m \psi_s}{2 J L_s}, & a_7 &= \frac{D}{J}, & a_8 &= \frac{n_p T_L}{J}, \\ c_1 &= a_3 u_{ds} + a_4 u_{dr}^0, & c_2 &= a_5 + a_3 u_{qs} + a_4 u_{qr}^0. \end{aligned} \quad (4)$$

Here u_{dr}^0, u_{qr}^0 are constant (open-loop) rotor voltages and $u = [u_1, u_2, u_3]^T$ is the control input: $u_{1,2} = a_4 \Delta u_{dr,qr}$ are the rotor-voltage actions of the rotor-side converter, and $u_3 = (n_p/J) \Delta T$ is a bounded auxiliary torque-correction channel (e.g., pitch-induced aerodynamic torque). Without control, $u = 0$.

Table I lists the nominal parameters of a representative 2-MW-class DFIG (stator-referred). They satisfy $L_s, L_r > L_m$ and $\sigma = 0.0661 \in (0, 1)$. The resulting coefficients are $a_1 = -31.14 \text{ s}^{-1}$, $a_2 = -1.013 \times 10^4 \text{ A/rad}$, $a_3 = -5.649 \times 10^3$ and $a_4 = 5.845 \times 10^3 \text{ A/(V}\cdot\text{s)}$, $a_5 = 1.018 \times 10^4 \text{ A/s}$, $a_6 = 8.19 \times 10^{-2} \text{ rad/(A}\cdot\text{s}^2)$ and $a_7 = 7.87 \times 10^{-6} \text{ s}^{-1}$. Because $u_{ds} = -\omega_s \psi_s$, $c_1 + a_2 \omega_s = a_4 u_{dr}^0$: the constant $a_3 u_{ds} = 3.18 \times 10^6 \text{ A/s}$ exactly compensates $a_2 \omega_s$, and $c_2 = 1.018 \times 10^4 + a_4 u_{qr}^0 \text{ A/s}$. Every constant of (3) is therefore obtained from Table I.

B. Dimensionless Model and Conditions for Chaos

Let $\tau = -1/a_1$, $\tilde{t} = t/\tau$, $k = a_7/(\tau a_6)$, and $z_1 = x_1/k$, $z_2 = x_2/k$, $z_3 = \tau(x_3 - \omega_s)$ (normalized slip speed). Then (3) with $u = 0$ becomes

$$\begin{cases} \dot{z}_1' = -z_1 - z_3 z_2 + \gamma z_3 + \varepsilon_1, \\ \dot{z}_2' = -z_2 + z_3 z_1 + \varepsilon_2, \\ \dot{z}_3' = s(z_1 - z_3) + \varepsilon_3, \end{cases} \quad (5)$$

TABLE I
NOMINAL PARAMETERS (2-MW-CLASS DFIG AND TURBINE)

Param.	Value	Param.	Value
R_s	2.6 mΩ	n_p	2
R_r	2.9 mΩ	V_s (line, rms)	690 V, 50 Hz
L_m	2.5 mH	ψ_s	1.793 Wb
$L_s = L_r$	2.587 mH	J	127 kg·m ²
σ	0.0661	D	10 ⁻³ N·m·s/rad
$\tau = -1/a_1$	32.1 ms	R, G	40 m, 80
ρ	1.225 kg/m ³	$\lambda_{opt}, C_{p,max}$	8.1, 0.48

where $(\cdot)' = d/d\tilde{t}$, $s = \tau D/J$, $\varepsilon_1 = \tau(c_1 + a_2 \omega_s)/k$, $\varepsilon_2 = \tau c_2/k$, $\varepsilon_3 = -\tau^2(a_7 \omega_s + a_8)$, and

$$\gamma = \frac{a_2 a_6 \tau}{a_7} = -\frac{3 n_p^2 L_m^2 \psi_s^2 \tau}{2 \sigma L_r L_s^2 D} < 0. \quad (6)$$

With physical parameters ($\sigma > 0$), γ is always negative. The divergence of (5) is $-(2+s) < 0$, so the system is dissipative.

For the nominal machine, $s = 2.5 \times 10^{-7}$ and the equilibrium is stable. Chaos appears only far from nominal. For $\gamma = -1$, $s = 8$, $\varepsilon_1 = \varepsilon_3 = 0$ and $\varepsilon_2 = -85$, the Lyapunov spectrum, computed with the QR-based variational method [38] over 500 time units, is (1.416, 0.003, -11.419). Its sum equals the divergence -10, and the Kaplan-Yorke dimension is 2.12. Fig. 1(a)–(b) shows the resulting double-scroll attractor. The bifurcation diagram and the largest Lyapunov exponent versus ε_2 (Fig. 1(c)–(d)) show a stable equilibrium for $\varepsilon_2 > -15$ and chaos for $\varepsilon_2 < -15$, interrupted by periodic windows near $\varepsilon_2 \approx -50.5$, -41.8 and -28.5.

Mapped back to Table I, this chaotic set requires $D/J \approx 250 \text{ s}^{-1}$ (e.g., $J \approx 13.6 \text{ kg}\cdot\text{m}^2$ and $D \approx 3.4 \text{ kN}\cdot\text{m}\cdot\text{s/rad}$) and a q -axis rotor voltage of several kilovolts. Chaos is therefore not a feature of normal operation of this machine but a severe fault/detuning scenario, for which the controller must still guarantee recovery. Following common practice [24], [25], chaos suppression is studied on (5), whereas MPPT tracking is studied in physical units.

IV. MPPT REFERENCE GENERATION

The wind speed is modelled as a mean value, three low-frequency harmonics and a smoothed Gaussian turbulence v_{turb} (first-order filter, time constant 0.5 s):

$$v(t) = V_{mean} + \sum_{k=1}^3 A_k \sin(2\pi f_k t + \phi_k) + v_{turb}(t), \quad (7)$$

with $V_{mean} = 9 \text{ m/s}$, $(A_k) = (1.3, 0.7, 0.4) \text{ m/s}$ and $(f_k) = (0.05, 0.13, 0.31) \text{ Hz}$ (Fig. 2).

The optimal speed follows the low-pass filtered wind $\bar{v}$ (time constant 2 s):

$$\omega_{r,ref}(t) = n_p G \lambda_{opt} \bar{v}(t)/R. \quad (8)$$

Since $T_e \propto i_{dr}$, the d -axis current reference is obtained from the torque balance of (3), so that it is consistent with $\omega_{r,ref}$:

$$i_{dr,ref} = \frac{\dot{\omega}_{r,ref} + a_7 \omega_{r,ref} + (n_p/J) T_L(v, \omega_{r,ref})}{a_6}. \quad (9)$$

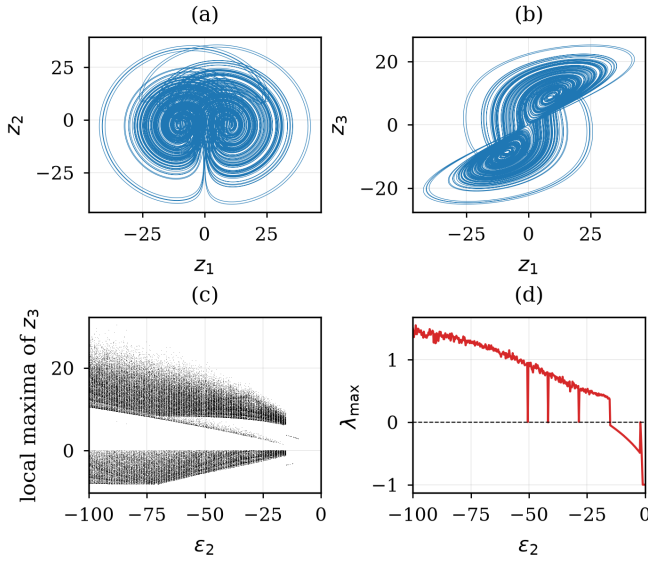


Fig. 1. Chaotic dynamics of (5) ($\gamma = -1$, $s = 8$, $\varepsilon_1 = \varepsilon_3 = 0$): (a) z_1 - z_2 and (b) z_1 - z_3 projections for $\varepsilon_2 = -85$; (c) bifurcation diagram (local maxima of z_3) and (d) largest Lyapunov exponent versus ε_2 .

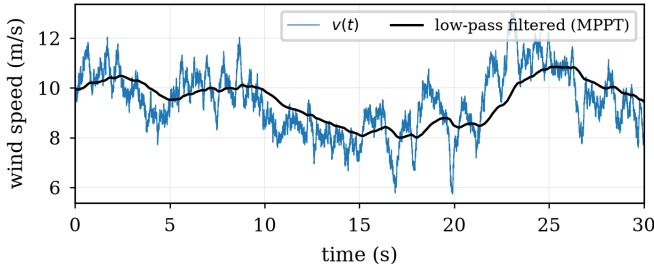


Fig. 2. Turbulent wind-speed profile and its low-pass filtered version used by the MPPT.

The stator reactive power is $Q_s = \frac{3}{2}V_s i_{qs} = \frac{3}{2}V_s(\psi_s - L_m i_{qr})/L_s$, so unity power factor ($Q_s = 0$) gives

$$i_{qr,ref} = \psi_s/L_m. \quad (10)$$

The generated stator active power and its reference are

$$P_s = -\frac{3}{2}V_s \frac{L_m}{L_s} i_{dr}, \quad P_{s,ref} = -\frac{3}{2}V_s \frac{L_m}{L_s} i_{dr,ref}. \quad (11)$$

The reference trajectory is $x_d = [i_{dr,ref}, i_{qr,ref}, \omega_{r,ref}]^T$ and the tracking error is $e = x - x_d$.

V. HYBRID FEEDBACK-LINEARIZATION/MPC DESIGN

A. Feedback Linearization

Write (3) as $\dot{x} = F(x, t) + u$ and let $\hat{F}$ be its nominal model. The control law

$$u = -\hat{F}(x, t) + \dot{x}_d + v \quad (12)$$

gives the error dynamics $\dot{e} = v + \Delta(x, t)$ with $\Delta = F - \hat{F}$ ($\Delta = 0$ nominally). This cancels the bilinear couplings ($\omega_s - x_3$) $x_{1,2}$ responsible for chaos, and leaves three decoupled integrators driven by the new input v .

B. Constrained MPC with Integral Action

For each channel i , the error is augmented with its integral, $\xi_i = [e_i, \int e_i dt]^T$, to remove steady-state errors due to Δ . Sampling with period T_s gives

$$\xi_i(k+1) = A\xi_i(k) + Bv_i(k), \quad A = \begin{bmatrix} 1 & 0 \\ T_s & 1 \end{bmatrix}, \quad B = \begin{bmatrix} T_s \\ T_s^2/2 \end{bmatrix}. \quad (13)$$

At each instant k , the MPC solves the quadratic program

$$\begin{aligned} \min_{v_i(\cdot|k)} \quad & \sum_{j=1}^N \|\xi_i(k+j|k)\|_Q^2 + \|\xi_i(k+N|k)\|_P^2 \\ & + r \sum_{j=0}^{N-1} v_i(k+j|k)^2 \\ \text{s.t.} \quad & (13), \quad \underline{v}_i(k) \leq v_i(k+j|k) \leq \bar{v}_i(k), \end{aligned} \quad (14)$$

where $Q = Q^T > 0$, $r > 0$, and the input bounds $\underline{v}_i = -\bar{u}_i - h_i$, $\bar{v}_i = \bar{u}_i - h_i$, with $h = -\hat{F} + \dot{x}_d$, enforce the actuator limit $|u_i| \leq \bar{u}_i$ on (12). Only the first optimal move is applied (receding horizon). The integral state is frozen when the limit is reached (conditional integration).

Theorem 1: Let $X = X^T > 0$ and Y satisfy the LMI

$$\begin{bmatrix} X & (AX + BY)^T & XQ^{1/2} & \sqrt{r}Y^T \\ AX + BY & X & 0 & 0 \\ Q^{1/2}X & 0 & I & 0 \\ \sqrt{r}Y & 0 & 0 & 1 \end{bmatrix} \succeq 0, \quad (15)$$

and set $P = X^{-1}$, $K = YX^{-1}$. Then

$$(A + BK)^T P (A + BK) - P + Q + rK^T K \preceq 0. \quad (16)$$

Let $\Omega = \{\xi : \xi^T P \xi \leq \alpha\}$, with $\alpha > 0$ small enough that $K\xi$ satisfies the input bounds in Ω . If $\Delta = 0$ and (14), augmented with the terminal constraint $\xi_i(k+N|k) \in \Omega$, is feasible at $k = 0$, then it remains feasible for all k and $\xi_i = 0$ is asymptotically stable, i.e., $x \rightarrow x_d$.

Proof: Applying the Schur complement to (15) with respect to its last three block rows gives $X - (AX + BY)^T X^{-1} (AX + BY) - XQX - rY^T Y \succeq 0$. Pre- and post-multiplying by $P = X^{-1}$ and substituting $Y = KX$ yields (16). Hence Ω is positively invariant under $v = K\xi$, and $\xi^T P \xi$ is a control Lyapunov function in Ω . By the standard terminal-cost/terminal-set argument [39], the optimal cost V_N^* of (14) then satisfies $V_N^*(k+1) - V_N^*(k) \leq -\|\xi_i(k)\|_Q^2 - r v_i(k)^2$ and the shifted solution completed by $K\xi$ is feasible, which proves recursive feasibility and asymptotic stability. ■

Remarks: (i) The maximal-volume solution of (15) satisfies (16) with equality, i.e., P is the solution of the discrete algebraic Riccati equation [40]. Consequently, when no constraint is active, (14) returns exactly $v_i = K\xi_i$. A single gain $K = [k_e, k_I]$, common to the three channels, is therefore obtained from the LMI and used in the implementation. (ii) Setting $k_I = 0$ and ignoring constraints recovers the pure feedback-linearization law $v = -k_e e$ ("FL only"), and applying $v = K\xi$ without the model terms $-\hat{F} + \dot{x}_d$ gives a PI law. These two baselines use the same gains as the

proposed controller, so the comparison isolates the effect of feedback linearization and of constrained MPC. (iii) The terminal constraint is not imposed explicitly in the simulations; convergence was verified a posteriori.

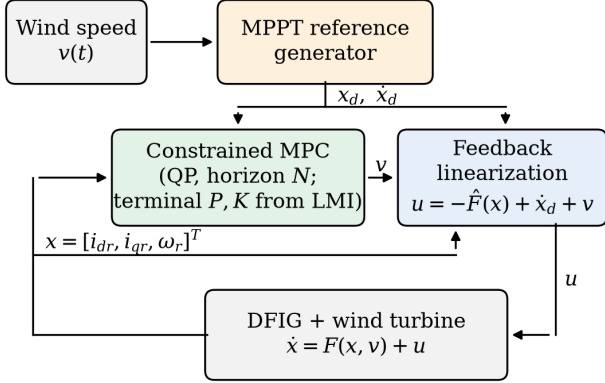


Fig. 3. Proposed hybrid feedback-linearization/MPC architecture.

The overall structure is shown in Fig. 3. The QP (14) has only N variables per channel and was solved by a bounded-variable least-squares method.

VI. SIMULATION RESULTS

All simulations integrate the plant with a fourth-order Runge–Kutta scheme (4–5 sub-steps per sampling period). The controller uses the nominal model; the plant uses the nominal or perturbed parameters.

A. Chaos Suppression

System (5) is started in the chaotic regime and the controller is switched on at $\tilde{t}_{on} = 6$ with $T_s = 0.01$, $N = 15$, $Q = \text{diag}(1, 10)$, $r = 10^{-4}$ and $|u_i| \leq 120$. The LMI gives $K = [-63.9, -192.4]$. The target $z^* = (2.02, -2.00, 2.02)$ corresponds to $\omega_r = 1.2\omega_s$ (super-synchronous operation with positive speed) and is sustained by $u_2^* = 78.9$ alone, which compensates the abnormal excitation ε_2 . Robustness is tested by scaling all coefficients of the plant by 0.8 and 1.2.

TABLE II
CHAOS SUPPRESSION PERFORMANCE (IAE OF $\|z - z^*\|$, SETTLING TIME t_s TO A 2% BAND, FINAL ERROR e_f ; NORMALIZED TIME)

Plant	Metric	FL–MPC	FL only	PI
nominal	IAE	0.37	0.31	0.69
	t_s	0.37	0.08	0.95
	e_f	0	0	0
–20 %	IAE	0.33	2.86	0.58
	t_s	0.39	–	0.89
	e_f	0	0.26	0
+20 %	IAE	1.51	3.94	1.87
	t_s	0.65	–	1.09
	e_f	0	0.26	0

Fig. 4(a) shows that the chaotic oscillations are suppressed as soon as the controller is activated, and the states settle at z^* . Table II and Fig. 4(b) compare the three controllers.

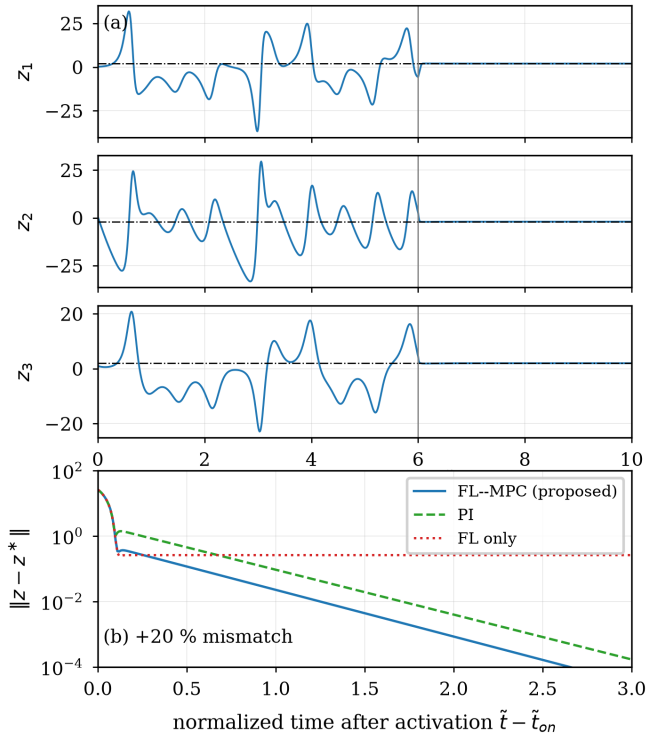


Fig. 4. Chaos suppression: (a) states of (5) with FL–MPC switched on at $\tilde{t} = 6$ (dash-dot: target z^*); (b) error norm after activation with +20 % plant mismatch.

With an exact model, FL only is slightly faster, because the integral action of FL–MPC adds a small tail. Under mismatch, however, FL only keeps a steady-state error (0.26), and PI, which does not cancel the nonlinear couplings, converges more slowly. FL–MPC converges without offset in all cases, with 19–47 % lower IAE than PI and settling times of 0.37–0.65 (12–21 ms with $\tau = 32.1$ ms). All controllers reach the input limit during the first instants; the MPC accounts for it explicitly.

B. MPPT Tracking of the 2-MW-Class DFIG

The model (3) with Table I is driven by the wind of Fig. 2 for 30 s, with $T_s = 1$ ms, $N = 10$, $Q = \text{diag}(1, 3)$ and $r = 10^{-6}$ (errors scaled by 1 kA, 1 kA and 100 rad/s). The limits are $|\Delta u_{dr}|, |\Delta u_{qr}| \leq 200$ V and $|\Delta T| \leq 2$ kN·m. Two mismatched plants are considered: M1 ($R_s, R_r + 20\%$, $L_m - 10\%$, $J + 20\%$) and M2 ($R_s, R_r - 20\%$, $L_m + 10\%$, $J - 20\%$), with the leakage inductances kept constant so that $L_s, L_r > L_m$ still holds.

Figs. 5 and 6 show that the rotor currents, the speed and the stator power follow the MPPT references despite turbulence. The speed stays in the admissible slip range (0.82 – $1.12\omega_s$), and P_s varies between 0.57 and 2.1 MW. The auxiliary torque channel is small (RMS 83 N·m, i.e., 0.7 % of rated torque), so the speed is governed by the generator torque. Table III shows that the power RMS error of FL–MPC stays below 0.9 % of rated power for all plants, five to eight times lower than with PI. For tracking, FL–MPC and FL only perform almost

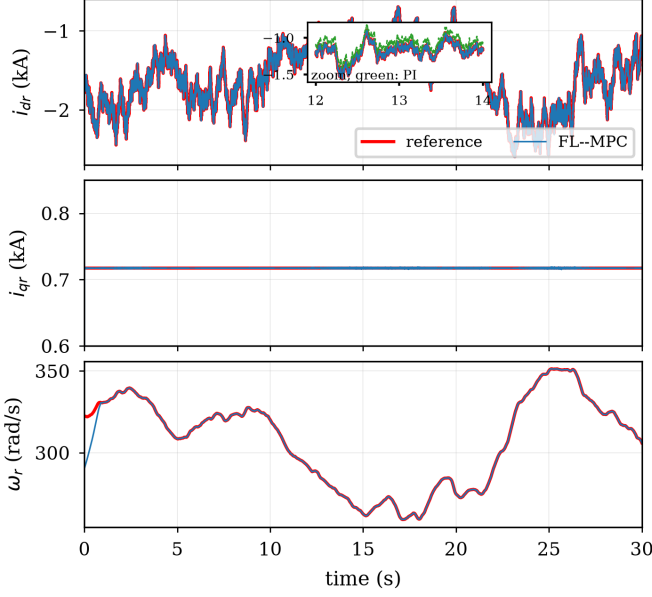


Fig. 5. Tracking of i_{dr} , i_{qr} and ω_r with FL-MPC (nominal plant). Inset: zoom including PI.

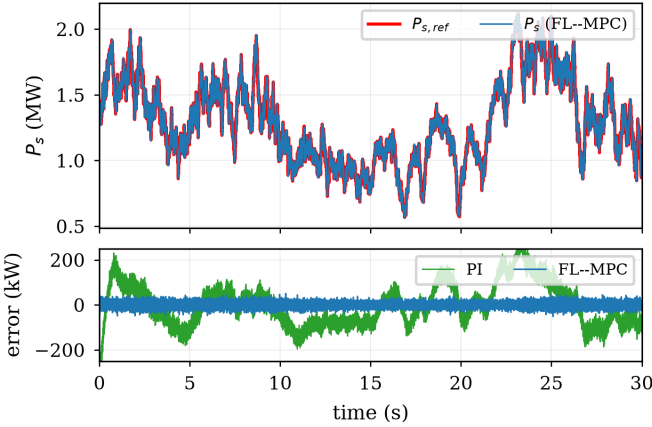


Fig. 6. Stator active power P_s and MPPT reference $P_{s,ref}$ (FL-MPC), and power tracking errors of FL-MPC and PI.

TABLE III
MPPT TRACKING PERFORMANCE ($t \geq 5$ s): RMS ERRORS OF P_s (% OF 2 MW), i_{dr} (A) AND ω_r (RAD/S)

Plant	RMSE	FL-MPC	FL only	PI
nominal	P_s	0.58	0.58	4.55
	i_{dr}	14.3	14.3	111.4
	ω_r	0.0012	0.0012	0.0069
M1	P_s	0.85	0.86	4.51
	i_{dr}	20.7	21.1	110.5
	ω_r	0.0044	0.0044	0.0070
M2	P_s	0.86	0.88	4.59
	i_{dr}	21.2	21.5	112.3
	ω_r	0.0063	0.0064	0.0085

identically (slight advantage of FL-MPC under mismatch), because the actuator limits are active only during the initial transient. The main benefits of the MPC are therefore its explicit, stability-preserving treatment of constraints and its offset-free behaviour, which matter in large transients such as the recovery from chaos.

VII. CONCLUSION

This paper revisited the nonlinear dynamics of a DFIG-based wind turbine with a physically consistent parameter set and explicit model constants. A dimensionless model shows that chaos, confirmed by a positive Lyapunov exponent and a bifurcation diagram, arises only under severely detuned damping and abnormal rotor excitation, i.e., as a fault scenario. A hybrid controller combining feedback linearization with a constrained MPC whose terminal ingredients are obtained from an LMI was proposed, with guaranteed nominal stability. Simulations show chaos suppression without steady-state error under $\pm 20\%$ mismatch, lower recovery IAE than PI, and MPPT power tracking with an RMS error below 0.9% of rated power. Future work will address converter and grid dynamics, low-voltage ride-through, and experimental validation.

CONFLICT OF INTEREST

The authors declare that there are no conflicts of interest.

REFERENCES

- [1] International Energy Agency, *World Energy Outlook 2024*. Paris, France: IEA, 2024. [Online]. Available: <https://www.iea.org/reports/world-energy-outlook-2024>
- [2] International Renewable Energy Agency, *Renewable Energy Statistics 2024*. Abu Dhabi, UAE: IRENA, Jul. 2024. [Online]. Available: <https://www.irena.org/publications/2024/Jul/Renewable-energy-statistics-2024>
- [3] United Nations Environment Programme, *Emissions Gap Report 2024*. Nairobi, Kenya: UNEP, 2024.
- [4] Global Wind Energy Council, *Global Wind Report 2024*. Brussels, Belgium: GWEC, Apr. 2024. [Online]. Available: <https://gvec.net/global-wind-report-2024/>
- [5] Global Wind Energy Council, *Global Wind Report 2025*. Brussels, Belgium: GWEC, Apr. 2025. [Online]. Available: <https://www.gvec.net/reports/globalwindreport/2025>
- [6] International Energy Agency, *Renewables 2024: Analysis and Forecast to 2030*. Paris, France: IEA, 2024. [Online]. Available: <https://www.iea.org/reports/renewables-2024>
- [7] R. Pena, J. C. Clare, and G. M. Asher, "Doubly fed induction generator using back-to-back PWM converters and its application to variable-speed wind-energy generation," *IEEE Proc.—Electr. Power Appl.*, vol. 143, no. 3, pp. 231–241, May 1996.
- [8] S. Müller, M. Deicke, and R. W. De Doncker, "Doubly fed induction generator systems for wind turbines," *IEEE Ind. Appl. Mag.*, vol. 8, no. 3, pp. 26–33, May/Jun. 2002.
- [9] S. Heier, *Grid Integration of Wind Energy: Onshore and Offshore Conversion Systems*, 3rd ed. Chichester, U.K.: Wiley, 2014.
- [10] M. G. Simões and F. A. Farret, *Modeling and Analysis with Induction Generators*, 3rd ed. Boca Raton, FL, USA: CRC Press, 2015.
- [11] A. A. Chhipa *et al.*, "Modeling and control strategy of wind energy conversion system with grid-connected doubly-fed induction generator," *Energies*, vol. 15, no. 18, Art. no. 6694, 2022, doi: 10.3390/en15186694.
- [12] L. Xu and Y. Wang, "Dynamic modeling and control of DFIG-based wind turbines under unbalanced network conditions," *IEEE Trans. Power Syst.*, vol. 22, no. 1, pp. 314–323, Feb. 2007, doi: 10.1109/TPWRS.2006.889113.

- [13] A. Tapia, G. Tapia, J. X. Ostolaza, and J. R. Sáenz, "Modeling and control of a wind turbine driven doubly fed induction generator," *IEEE Trans. Energy Convers.*, vol. 18, no. 2, pp. 194–204, Jun. 2003, doi: 10.1109/TEC.2003.811727.
- [14] Y. Lei, A. Mullane, G. Lightbody, and R. Yacamini, "Modeling of the wind turbine with a doubly fed induction generator for grid integration studies," *IEEE Trans. Energy Convers.*, vol. 21, no. 1, pp. 257–264, Mar. 2006, doi: 10.1109/TEC.2005.847958.
- [15] H. Nian, Y. Song, P. Zhou, and Y. He, "Improved direct power control of a wind turbine driven doubly fed induction generator during transient grid voltage unbalance," *IEEE Trans. Energy Convers.*, vol. 26, no. 3, pp. 976–986, Sep. 2011, doi: 10.1109/TEC.2011.2158436.
- [16] D. Xiang, L. Ran, P. J. Tavner, and S. Yang, "Control of a doubly fed induction generator in a wind turbine during grid fault ride-through," *IEEE Trans. Energy Convers.*, vol. 21, no. 3, pp. 652–662, Sep. 2006.
- [17] J. Morren and S. W. H. de Haan, "Ridethrough of wind turbines with doubly-fed induction generator during a voltage dip," *IEEE Trans. Energy Convers.*, vol. 20, no. 2, pp. 435–441, Jun. 2005, doi: 10.1109/TEC.2005.845526.
- [18] M. Rahimi and M. Parniani, "Efficient control scheme of wind turbines with doubly fed induction generators for low-voltage ride-through capability enhancement," *IET Renew. Power Gener.*, vol. 4, no. 3, pp. 242–252, May 2010, doi: 10.1049/iet-rpg.2009.0072.
- [19] M. Tsili and S. Papathanassiou, "A review of grid code technical requirements for wind farms," *IET Renew. Power Gener.*, vol. 3, no. 3, pp. 308–332, Sep. 2009, doi: 10.1049/iet-rpg.2008.0070.
- [20] M. Rahimi and M. Parniani, "Dynamic behavior and transient stability analysis of fixed speed wind turbines," *Renew. Energy*, vol. 34, no. 12, pp. 2613–2624, Dec. 2009, doi: 10.1016/j.renene.2009.06.019.
- [21] Y. Yu, Z. Mi, and X. Liu, "Analysis of chaos in doubly fed induction generator and sliding mode control of chaos synchronization," *Acta Phys. Sin.*, vol. 60, no. 7, Art. no. 070509, 2011, doi: 10.7498/aps.60.070509.
- [22] C. N. Van, N. T. Hai, and N. P. Quang, "Chaos control of doubly-fed induction generator via delayed feedback control," *Eng. Technol. Appl. Sci. Res.*, vol. 13, no. 2, pp. 10588–10594, Apr. 2023, doi: 10.48084/etasr.5812.
- [23] P. A. Meehan, "Prediction and suppression of chaos following flutter in wind turbines," *Nonlinear Dyn.*, vol. 111, pp. 22153–22176, 2023, doi: 10.1007/s11071-023-08841-9.
- [24] H. Ren and D. Liu, "Nonlinear feedback control of chaos in permanent magnet synchronous motor," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 53, no. 1, pp. 45–50, Jan. 2006.
- [25] M. Messadi and A. Mellit, "Control of chaos in an induction motor system with LMI predictive control and experimental circuit validation," *Chaos, Solitons Fractals*, vol. 97, pp. 51–58, 2017.
- [26] G. Chen, "A simple adaptive feedback control method for chaos and hyper-chaos control," *Appl. Math. Comput.*, vol. 217, no. 17, pp. 7258–7264, 2011.
- [27] H. Takhi, K. Kemih, L. Moysis, *et al.*, "Passivity based control and synchronization of perturbed uncertain chaotic systems and their microcontroller implementation," *Int. J. Dyn. Control*, vol. 8, no. 3, pp. 973–990, 2020, doi: 10.1007/s40435-020-00618-x.
- [28] A. Boukabou, A. Chebbah, and N. Mansouri, "Predictive control of continuous chaotic systems," *Int. J. Bifurcation Chaos*, vol. 18, no. 2, pp. 587–592, 2008, doi: 10.1142/S0218127408020501.
- [29] A. Senouci and A. Boukabou, "Predictive control and synchronization of chaotic and hyperchaotic systems based on a T-S fuzzy model," *Math. Comput. Simul.*, vol. 105, pp. 62–78, 2014.
- [30] M. Messadi and A. Mellit, "Predictive control of a chaotic permanent magnet synchronous generator in a wind turbine system," *Chin. Phys. B*, vol. 24, no. 1, Art. no. 010502, 2015.
- [31] H. Takhi, L. Moysis, N. Machkour, C. Volos, K. Kemih, and M. Ghanes, "Predictive control and synchronization of uncertain perturbed chaotic permanent-magnet synchronous generator and its microcontroller implementation," *Eur. Phys. J. Spec. Top.*, vol. 231, no. 3, pp. 443–451, 2022, doi: 10.1140/epjs/s11734-021-00422-4.
- [32] A. Mechter, K. Kemih, and M. Ghanes, "Sliding mode control of a wind turbine with exponential reaching law," *Acta Polytech. Hung.*, vol. 12, no. 3, pp. 167–183, 2015.
- [33] I. Bouguettah, M. Messadi, K. Kemih, A. T. Azar, and A. R. Mahlous, "Adaptive passive fault tolerant control of DFIG-based wind turbine using a self-tuning fractional integral sliding mode control," *Front. Energy Res.*, vol. 12, Art. no. 1429877, 2024, doi: 10.3389/fenrg.2024.1429877.
- [34] M. Messadi, K. Kemih, and H. Hamiche, "A new secure communications scheme based on synchronization of hybrid chaotic system," *Chaos Theory Appl.*, vol. 5, no. 3, pp. 160–166, 2023, doi: 10.51537/chaos.1276714.
- [35] P. Jiang, T. Zhang, J. Geng, P. Wang, and L. Fu, "An MPPT strategy for wind turbines combining feedback linearization and model predictive control," *Energies*, vol. 16, no. 10, Art. no. 4244, 2023, doi: 10.3390/en16104244.
- [36] S. Regani, M. Messadi, and K. Kemih, "Modeling and nonlinear dynamic analysis of a DFIG-based wind turbine system," submitted for publication.
- [37] A. Mechter, K. Kemih, and M. Ghanes, "Backstepping control of a wind turbine for low wind speeds," *Nonlinear Dyn.*, vol. 84, no. 4, pp. 2435–2445, 2016, doi: 10.1007/s11071-016-2655-y.
- [38] A. Wolf, J. B. Swift, H. L. Swinney, and J. A. Vastano, "Determining Lyapunov exponents from a time series," *Physica D*, vol. 16, no. 3, pp. 285–317, 1985.
- [39] D. Q. Mayne, J. B. Rawlings, C. V. Rao, and P. O. M. Scokaert, "Constrained model predictive control: Stability and optimality," *Automatica*, vol. 36, no. 6, pp. 789–814, 2000, doi: 10.1016/S0005-1098(99)00214-9.
- [40] M. V. Kothare, V. Balakrishnan, and M. Morari, "Robust constrained model predictive control using linear matrix inequalities," *Automatica*, vol. 32, no. 10, pp. 1361–1379, 1996.